

Name: _____ Partner: _____ Period: _____

Part 1 — Graphing $x^2 + y^2 = 36$

1. Substitute each
- x
- value into
- $x^2 + y^2 = 36$
- and solve for both
- y
- values.

x	y (positive / upper)	y (negative / lower)
-6	0	0
$-3\sqrt{2}$	$3\sqrt{2} \approx 4.243$	$-3\sqrt{2} \approx -4.243$
0	6	-6
$3\sqrt{2}$	$3\sqrt{2} \approx 4.243$	$-3\sqrt{2} \approx -4.243$
6	0	0

2. This equation defines
- 2**
- explicit function(s).

$$f_1(x) = \sqrt{36 - x^2} \quad f_2(x) = -\sqrt{36 - x^2}$$

3. Domain of
- f_1
- :
- $[-6, 6]$
- Domain of
- f_2
- :
- $[-6, 6]$

Part 2 — Analyzing Co-variation on $x^2 + y^2 = 36$

x	y
3.0	5.196
3.1	5.155
3.2	5.113

- 4.
- $\Delta y / \Delta x$
- from
- $x = 3.0$
- to
- $x = 3.1$
- :

$$\frac{\Delta y}{\Delta x} = \frac{5.155 - 5.196}{3.1 - 3.0} = -0.41$$

- 5.
- $\Delta y / \Delta x$
- from
- $x = 3.1$
- to
- $x = 3.2$
- :

$$\frac{\Delta y}{\Delta x} = \frac{5.113 - 5.155}{3.2 - 3.1} = -0.42$$

6. Is
- $\Delta y / \Delta x$
- positive or negative?
- Negative (both times).**

7. What does this tell you?

As x increases, y decreases along the upper semicircle — opposite-direction co-variation.

Part 3 — The Equation $xy = 12$

8. $y = 12/x$

9.

x	y
1	12
2	6
3	4
4	3
6	2

10. As x increases (for $x > 0$), y **decreases**.

11.

$$\frac{\Delta y}{\Delta x} = \frac{4 - 6}{3 - 2} = -2$$

Positive or negative? **Negative**. What does this confirm?

Co-variation is negative: as x increases, y decreases. The variables change in opposite directions.

Teacher Note

Part 1: For $x = -3\sqrt{2} \approx -4.243$: $(-3\sqrt{2})^2 = 18$, so $y^2 = 18$, giving $y = \pm 3\sqrt{2}$. Students may need a reminder to square the expression first.

Part 2: The ratios -0.41 and -0.42 are approximations; accept answers close to these values. The key insight is that both are negative, confirming opposite-direction co-variation on the upper semicircle.

Part 3: $\Delta y/\Delta x = (4 - 6)/(3 - 2) = -2$. Students should connect the negative ratio to the fact that $xy = 12$ means one variable must decrease if the other increases (their product stays fixed).